

Problem 6 (a)

$$\text{In[30]:= } E_x' = E_x$$

$$B_x' = B_x$$

$$E_y' = \gamma * (E_y - v * B_z)$$

$$B_y' = \gamma * (B_y + (v / c^2) * E_z)$$

$$E_z' = \gamma * (E_z + v * B_y)$$

$$B_z' = \gamma * (B_z - (v / c^2) * E_y)$$

$$\gamma = 1 / \text{Sqrt}[1 - (v^2 / c^2)]$$

$$\text{Out[30]= } E_x$$

$$\text{Out[31]= } B_x$$

$$\text{Out[32]= } \frac{E_y - B_z v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[33]= } \frac{B_y + \frac{E_z v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[34]= } \frac{E_z + B_y v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[35]= } \frac{B_z - \frac{E_y v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[36]} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Below we compute the expanded form of $\vec{E} \cdot \vec{B}$

$$\text{In[39]} := \mathbf{Ex' * Bx' + Ey' * By' + Ez' * Bz'}$$

$$\text{Out[39]} = Bx Ex + \frac{\left(Ez + By v\right) \left(Bz - \frac{Ey v}{c^2}\right)}{1 - \frac{v^2}{c^2}} + \frac{\left(Ey - Bz v\right) \left(By + \frac{Ez v}{c^2}\right)}{1 - \frac{v^2}{c^2}}$$

$$\text{In[38]} := \mathbf{FullSimplify[Ex' * Bx' + Ey' * By' + Ez' * Bz']}$$

$$\text{Out[38]} = Bx Ex + By Ey + Bz Ez$$

We verified $E \cdot B$ is invariant.

Problem 6 (b)

First let's compute $E'^2 - c^2 B'^2$

$$\text{In[41]} := \left(\mathbf{Ex' * Ex' + Ey' * Ey' + Ez' * Ez'} \right) -$$

$$c^2 \left(\mathbf{Bx' * Bx' + By' * By' + Bz' * Bz'} \right)$$

$$\text{Out[41]} = Ex^2 + \frac{\left(Ez + By v\right)^2}{1 - \frac{v^2}{c^2}} + \frac{\left(Ey - Bz v\right)^2}{1 - \frac{v^2}{c^2}} - c^2 \left(Bx^2 + \frac{\left(Bz - \frac{Ey v}{c^2}\right)^2}{1 - \frac{v^2}{c^2}} + \frac{\left(By + \frac{Ez v}{c^2}\right)^2}{1 - \frac{v^2}{c^2}} \right)$$

$$\text{In[40]} := \mathbf{FullSimplify\left[\left(Ex' * Ex' + Ey' * Ey' + Ez' * Ez'\right) -}$$

$$c^2 \left(Bx' * Bx' + By' * By' + Bz' * Bz' \right) \right]$$

$$\text{Out[40]} = -\left((Bx^2 + By^2 + Bz^2) c^2 \right) + Ex^2 + Ey^2 + Ez^2$$

Full Simplification shows that we have gotten the non-prime factors of E_i , B_j intact, that means that the spacetime-interval like quantity $E^2 - c^2 B^2$ is invariant.

Problem 6 (c)

Answer: No

Reason: For one system if $B=0$ then $S^2 = E^2 - c^2 B^2 = E^2 > 0$ which implies $S^2 > 0$

But now for another system where $E = 0$ we get $S^2 = -c^2 B^2$ which implies $S^2 < 0$

But we know that S^2 is conserved for all frames, hence it can't be greater and lower than zero for the same system in two different frames. This means the answer to the question is NO.